Code

Survival Model Selection and Validation for HEOR in R

A practical workflow for fitting, diagnosing, comparing, and justifying parametric survival models

Xin (David) Zhao

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Introduction

In health economic evaluations such as cost-effectiveness analyses, estimating the long-term health benefits of innovative treatments over a lifetime horizon is essential. However, randomized controlled trials (RCTs) typically have short follow-up periods. Therefore, following health technology assessment (HTA) guidelines, it is important to extrapolate survival data in a statistically robust and clinically justified manner to predict life expectancy for both new treatments and standard care.

This R notebook provides a reusable, HTA-oriented framework for fitting survival models in R and for evaluating, interpreting, and defending the model selected for economic evaluation.

Comprehensive workflow and R tools

NICE-recommended survival extrapolation workflow

NICE TSD14 recommends a structured four-step procedure to ensure that survival extrapolations are both statistically sound and clinically credible.

Each step builds on the last, guiding the analyst from exploratory data inspection through to a fully justified model selection that can withstand scrutiny in a health technology appraisal.

Step 1 — Assess model assumptions & hazard structure

Examine the observed data with graphical and statistical tools to understand which survival and hazard shapes are plausible before selecting a model.

Step 2 — Fit candidate models

Fit the full standard parametric family (Exponential, Weibull, Gompertz, Log-normal, Log-logistic, Generalized Gamma) and flexible alternatives where justified.

Step 3 — Evaluate statistical fit & graphical diagnostics

Compare fitted curves against the Kaplan-Meier estimate using AIC/BIC and residual diagnostics. Visually inspect Q-Q plots and Cox-Snell residual plots to detect systematic departures.

Step 4 — Evaluate external validity & decision uncertainty

Assess whether the extrapolated curves are clinically plausible, consistent with external evidence, and robust to alternative model choices.

Step 5 - Model selection and structural uncertainty

Select the most appropriate survival model based on statistical fit, graphical diagnostics, clinical plausibility, and external validation. Consider alternative models to assess structural uncertainty and its impact on decision-making.

R packages

Three specialized R packages were used to implement the workflow:

- `survival`: Fit Kaplan-Meier curves and create visualizations (e.g., cumulative hazard plots)
- `flexsurv`: Fit parametric and flexible survival models, perform residual diagnostics, and create visualizations
- `GofCens`: Assess goodness of fit and create visualizations

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```
# Install required packages
required_pkgs <- c("survival", "flexsurv", "GofCens")

for (pkg in required_pkgs) {
    if (!requireNamespace(pkg, quietly = TRUE)) {
        install.packages(pkg)
}
}
```

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```
# Load required packages

invisible(lapply(required_pkgs, library, character.only = TRUE))
```

Input data

The input data, censored time-to-event data (locoregional control), were digitized from an RCT by Bonner et al., 2006, which investigated the effect of cetuximab treatment on patients with head and neck cancer.

The `IPD_ctrl_gp.csv` and `IPD_treat_gp.csv` files contain time-to-event data for the control (n=213) and treatment (n=211) arms, with a 50-month follow-up.

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```
# Control arm
ipd_ctrl <- read.csv("IPD_ctrl_gp.csv")

# Treatment arm
ipd_trt <- read.csv("IPD_treat_gp.csv")

# Merge the data sets from the two arms
ipd_merged <- rbind(ipd_ctrl, ipd_trt)
```

Inspect the first rows of the combined dataset. The columns are defined as follows:

- `time`: Follow-up time in months.
- `status`: 1 indicates an event; 0 indicates censoring.
- `treat`: 1 indicates treatment; 0 indicates control.

Hide

```
# Inspect the first rows of the merged dataset
head(ipd_merged, 15)
```

```
##       time status treat
## 1  0.00274      1     0
## 2  0.00274      1     0
## 3  0.26100      1     0
## 4  0.77900      1     0
## 5  0.77900      1     0
## 6  0.77900      1     0
## 7  1.03450      0     0
## 8  1.29000      1     0
## 9  1.81000      1     0
## 10 2.07000      1     0
## 11 2.33000      0     0
## 12 2.59000      1     0
## 13 2.59000      1     0
## 14 2.85000      1     0
## 15 2.85000      1     0
```

Step 1 - Assess model assumptions & hazard structure

Use graphical and statistical tools to determine which survival and hazard shapes are plausible before selecting a model.

Step 1.1 Log cumulative hazard plot

A (log) cumulative hazard plot is recommended to answer key questions:

- What hazard structure is supported?
- Are proportional hazards (PH) reasonable?

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```
# Fit Kaplan-Meier curves, stratified by treatment groups
surv_km <- survival::survfit(Surv(time, status) ~ treat,
  data = ipd_merged,
  type = "kaplan-meier"
)
```

To assess whether hazards are constant over time (e.g., exponential distribution), plot cumulative hazard versus time. To evaluate monotonic changes in hazard (e.g., Weibull and Gompertz distributions), plot the logarithm of cumulative hazard against the logarithm of time.

Hide

```
# Plot cumulative hazard
old_par <- par(no.readonly = TRUE)

par(mfrow = c(1, 2)) # Set up a 1x2 plotting area

plot(surv_km,
  fun = "cumhaz",
  xlab = "Time (months)",
  ylab = "Cumulative hazard",
  col = c("black", "green"),
  main = "Cumulative hazard plot"
)

legend("topleft",
  legend = c("Control", "Treatment"),
  col = c("black", "green"),
  lty = 1,
  lwd = 1,
  bty = "n"
)

# Use logarithmic scales for both axes
plot(surv_km,
  fun = "cumhaz",
  log = "xy",
```

```
  xlab = "Log, Time (months)",
  ylab = "Log, Cumulative hazard",
  col = c("black", "green"),
  main = "Log cumulative hazard plot"
)

legend("topleft",
  legend = c("Control", "Treatment"),
  col = c("black", "green"),
  lty = 1,
  lwd = 1,
  bty = "n"
)
```

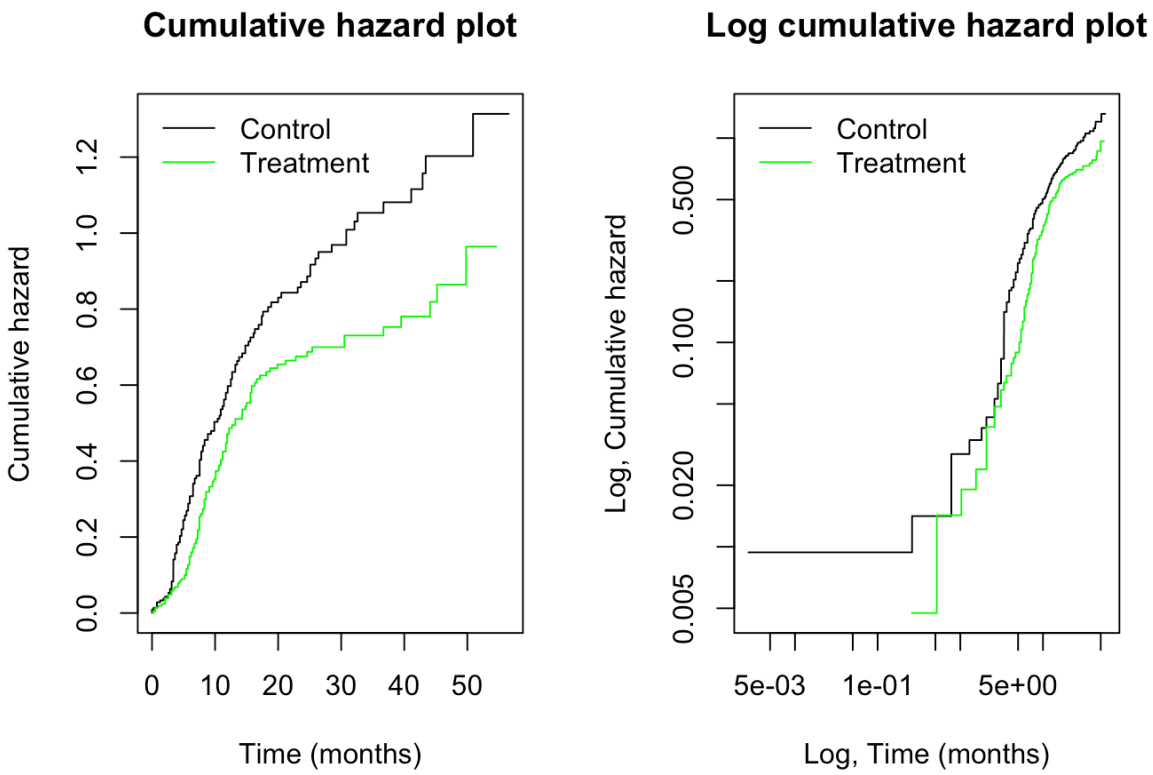


Figure 1. Log cumulative hazard plots for the control and treatment arms.

The cumulative hazard curves (left) change slope around 10–20 months, suggesting that the hazard may vary over follow-up. A constant-hazard exponential model therefore appears questionable over the full period. Piecewise or more flexible models may be worth considering.

On the log-cumulative-hazard scale (right), approximate linearity over a sustained interval would support further consideration of Weibull-type models. Examine the apparent early and later departures from linearity before concluding that one distribution is suitable. Evaluate Gompertz and other families using appropriate distribution-specific diagnostics.

The treatment-arm curves (right) appear approximately parallel over parts of follow-up, but early instability and possible time-varying separation mean that PH is not clearly established. Consider PH provisional and

assess it alongside suitable residual-based tests and clinical expertise. If PH is not adequately supported, consider arm-specific or time-varying approaches.

Step 2. Fit common parametric survival models to both arms

Fit the same candidate distribution type to both arms, allowing arm-specific scale and, where appropriate, shape parameters. Use a treatment-covariate PH parameterization only if PH is adequately supported. If you select different distribution families for the two arms, provide strong statistical, biological, and clinical justification.

Hide

```
model_distr_list <- c(
  "exponential",
  "weibull",
  "gompertz",
  "lognormal",
  "llogis",
  "gengamma"
)

fitted_models <- lapply(model_distr_list, function(distr) {
  flexsurv::flexsurvreg(
    Surv(time, status) ~ treat,
    data = ipd_merged,
    dist = distr
  )
})

names(fitted_models) <- model_distr_list
```

Step 3.Evaluate statistical fit & graphical diagnostics

Step 3.1 Evaluate the AIC of fitted models

Evaluate the relative fit of models fitted to the same observed data, accounting for model complexity, using the Akaike information criterion (AIC).

AIC does not establish absolute adequacy, robustness or the credibility of the extrapolated tail; those questions require graphical diagnostics, assessment of censoring, clinical plausibility and external validation.

Hide

```
aic_values <- sapply(fitted_models, function(model) {
  model$AIC
})
```

```
# Create a data frame to display AIC values for each distribution
aic_df <- data.frame(
  Distribution = names(aic_values),
  AIC = aic_values
)

# Drop row names
rownames(aic_df) <- NULL

# Sort the data frame by AIC values in ascending order
aic_df <- aic_df[order(aic_df$AIC), ]

aic_df # gompertz has the lowest AIC value
```

```
##   Distribution      AIC
## 3     gompertz 2173.631
## 5       llogis 2179.306
## 6     gengamma 2191.250
## 4    lognormal 2198.150
## 2      weibull 2204.574
## 1  exponential 2215.113
```

Among the fitted candidates, Gompertz has the lowest AIC and therefore the strongest relative support for fit to the observed data after accounting for model complexity.

Step 3.2 Compare KM curves with fitted models

As a complementary evaluation, we visually inspect the fitted model curves by comparing them with nonparametric Kaplan-Meier curves for the control and treatment arms separately.

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```
par(mfrow = c(6, 2),
    cex.main = 2,
    cex.axis = 1.5,
    cex.lab = 1.5)


for (mod in names(fitted_models)) {

  par(mfrow = c(1, 2))

  # Control arm

  km_ctrl <- survfit(Surv(time, status) ~ 1,
  data = ipd_merged[ipd_merged$treat == 0, ])

  plot(
    km_ctrl,
    main = paste("Control arm: KM vs", mod),
    xlab = "Time",
```

```
  ylab = "Survival probability",
  lty = 1,
  col = "black",
  mark.time = FALSE
)

lines(
  fitted_models[[mod]],
  newdata = data.frame(treat = 0),
  type = "survival",
  ci = FALSE,
  lty = 1,
  col = "blue"
)

# Treatment arm

km_trt <- survfit(Surv(time, status) ~ 1,
data = ipd_merged[ipd_merged$treat == 1, ])

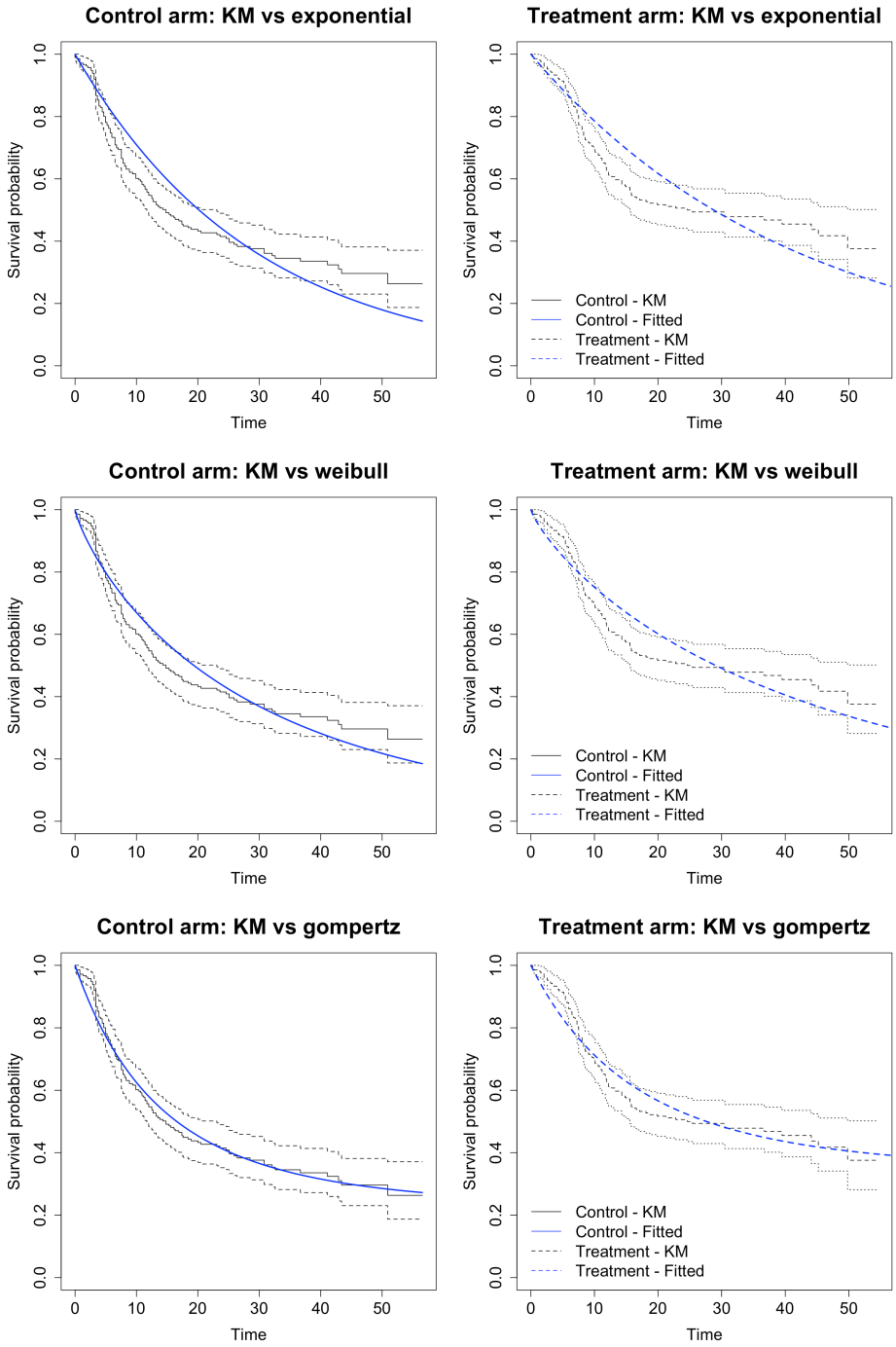
plot(
  km_trt,
  main = paste("Treatment arm: KM vs", mod),
  xlab = "Time",
  ylab = "Survival probability",
  lty = 2,
  mark.time = FALSE,
  col = "black"
)

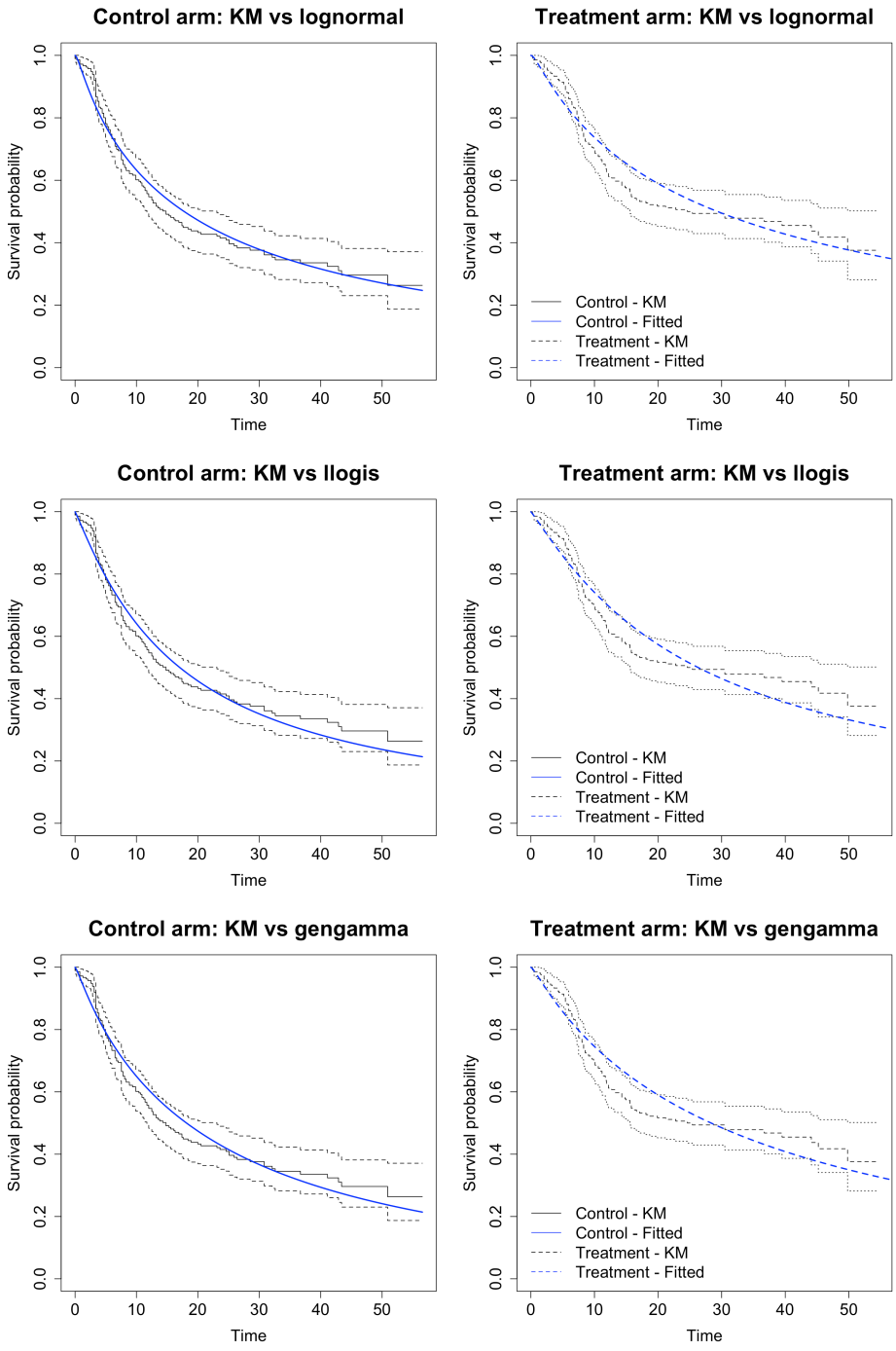
lines(
  fitted_models[[mod]],
  newdata = data.frame(treat = 1),
  type = "survival",
  ci = FALSE,
  lty = 2,
  col = "blue"
)

legend(
  "bottomleft",
  legend = c(
    "Control - KM",
    "Control - Fitted",
    "Treatment - KM",
    "Treatment - Fitted"
  ),
  col = c("black", "blue", "black", "blue"),
  lty = c(1, 1, 2, 2),
  lwd = 1,
  bty = "n",
  cex = 1.5
```

```
  )

}
```


Control arm: KM vs exponential
Treatment arm: KM vs exponential
Control arm: KM vs weibull
Treatment arm: KM vs weibull
Control arm: KM vs gompertz
Treatment arm: KM vs gompertz
Survival probability
Time
0.0
0.2
0.4
0.6
0.8
1.0
0
10
20
30
40
50
Control - KM
Control - Fitted
Treatment - KM
Treatment - Fitted



Gompertz has favorable AIC and generally acceptable graphical fit within the observed period, but log-normal and log-logistic also remain plausible on some diagnostics. The evidence supports retaining a set of candidate models rather than declaring Gompertz uniquely best. Selecting a base-case model should also reflect the completeness of follow-up and the clinical plausibility of each extrapolated tail.

Step 3.3 Q-Q plots for right-censored data

The purpose of using Q-Q plots is to determine whether specific distribution families are compatible with the observed data.

Below are the Q-Q plots for the control arm. Similar plots can be generated for the treatment arm to assess the fit of candidate distributions across both study groups.

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```
# Define a vector of recommended parametric models for loop iterations
distr_list <- c(
  "exponential",
  "weibull",
  "gompertz",
  "lognormal",
  'loglogistic',
  "llogis",
  "gengamma"
)

for (distr in distr_list) {

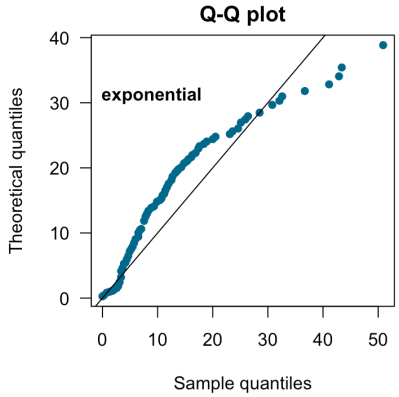
  tryCatch(
    {

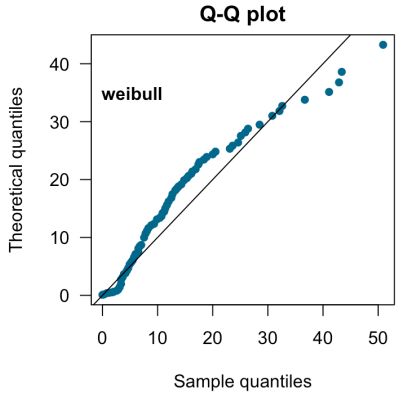
    GofCens::probPlot(
        ipd_ctrl$time,
        ipd_ctrl$status,
        distr = distr,
        plots = "QQ",
        mtitle = FALSE,
        ggplo = FALSE,
        prnt = FALSE
      )

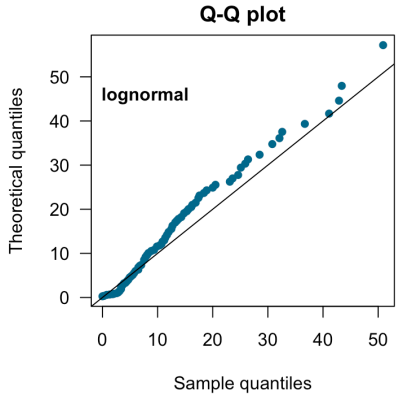
    usr <- par("usr")

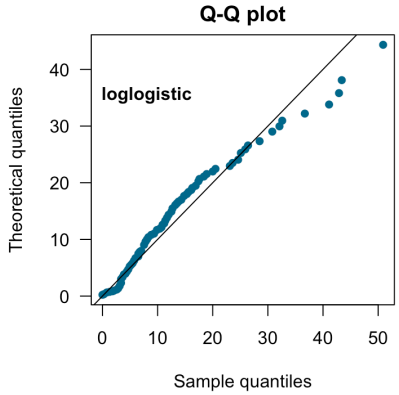
    text(x = usr[1] + 0.10 * (usr[2] - usr[1]),
         y = usr[4] - 0.12 * (usr[4] - usr[3]),
         labels = distr,
         adj = c(0, 1),
         font = 2,
         cex = 1)

    },
    error = function(error_message) {
      warning(
        "Distribution '", distr,
        "' is not available for GofCens::probPlot(): ",
        conditionMessage(error_message)
      )
    }
  )
}
```









The censored Q–Q plots show departures from the reference line for each evaluated family, particularly at later quantiles where information may be sparse. These plots do not clearly identify a single preferred distribution and should be treated as screening evidence alongside arm-specific hazard plots, residual

diagnostics, AIC/BIC, and clinical plausibility. This implementation cannot evaluate Gompertz due to software limitations, so assess it using other diagnostics.

Step 3.4 Residual plots (Cox-Snell)

Residual plots can detect systematic lack of fit in candidate models.

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```
par(
  mfrow = c(2, 6),
  oma = c(4, 4, 2, 1),   # outer margins
  mar = c(3, 2, 4.5, 1)    # individual plot margins
)

# Define a list of datasets for the two arms for loop iterations
data_list  <- list(
  ipd_ctrl = ipd_ctrl,
  ipd_trt = ipd_trt
)

for (data_name in names(data_list)) {

  data <- data_list[[data_name]]

  for (distr in distr_list) {

    tryCatch(
      {
        # Fit parametric survival model
        fit <- flexsurv::flexsurvreg(
          survival::Surv(time, status) ~ 1,
          data = data,
          dist = distr
        )

        # Cox-Snell residuals
        cs <- flexsurv::coxsnell_flexsurvreg(fit)

        # Estimate cumulative hazard of residuals
        surv <- survival::survfit(
          survival::Surv(cs$est, data$status) ~ 1
        )

        # Plot
        plot(
          surv,
          fun = "cumhaz",
          main = distr,
          xlab = "",
          ylab = ""
        )
```

```
        # Ideal Cox-Snell reference line
        abline(0, 1, col = "red")

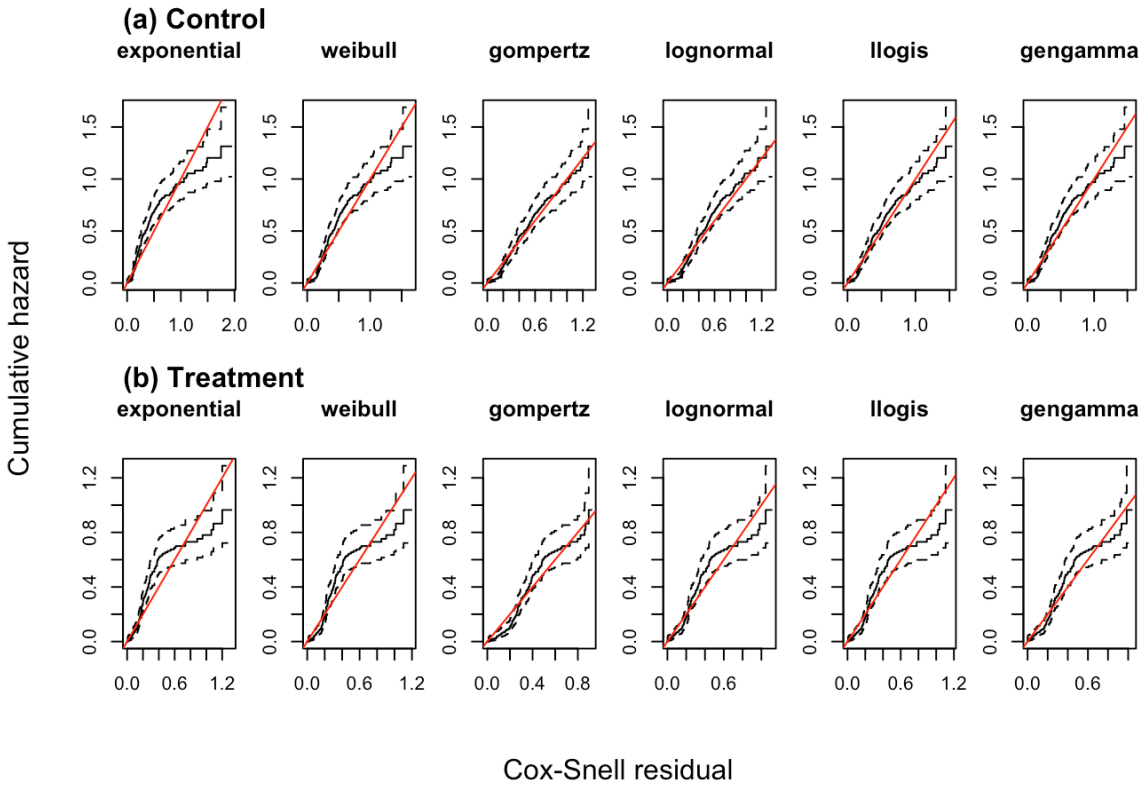
        if (distr == distr_list[1]) {

          panel_label <- if (data_name == names(data_list)[1]) {
            "(a) Control"
          } else {
            "(b) Treatment"
          }

          mtext(
            panel_label,
            side = 3,
            line = 3,
            adj = 0,
            font = 2,
            cex = 1
          )
}
      },
      error = function(error_message) {
        warning(
          "Model failed for dataset '", data_name,
          "' with distribution '", distr, "': ",
          conditionMessage(error_message)
        )
      }
    )
  }
}


# Global axis labels
mtext(
  "Cox-Snell residual",
  side = 1,
  outer = TRUE,
  line = 2
)

mtext(
  "Cumulative hazard",
  side = 2,
  outer = TRUE,
  line = 2
)
```



For a well-fitting model, the estimated cumulative hazard of the Cox–Snell residuals should lie approximately along the 45° line. Several candidates track the line over much of the observed range, with differing localized departures and increasing uncertainty toward the tail. These plots provide no definitive basis for selecting a particular distribution on their own and should be interpreted with the other diagnostics.

Step 4. Evaluate external validity & decision uncertainty

To evaluate impact of different survival models on extrapolated survival estimates, we generate long-term survival predictions for each fitted model and compare them graphically. Assume constant treatment effect beyond the observed follow-up period.

Hide

```
require(tidyr)
require(dplyr)
require(ggplot2)


times <- 0:120

pr_long <- dplyr::data_frame()

for (mod in names(fitted_models)) {

  pr <- summary(fitted_models[[mod]],
                newdata = data.frame(treat = c(0, 1)),
```

```
              type = "survival",
              t = times)

  pr_control <- as.data.frame(pr[[1]]) |>
    dplyr::mutate(group = "Control") |>
    dplyr::mutate(model = mod)

  pr_treatment <- as.data.frame(pr[[2]]) |>
    dplyr::mutate(group = "Treatment") |>
    dplyr::mutate(model = mod)

  pr_long <- dplyr::bind_rows(
    pr_long,
    pr_control,
    pr_treatment
)

}
```

Hide

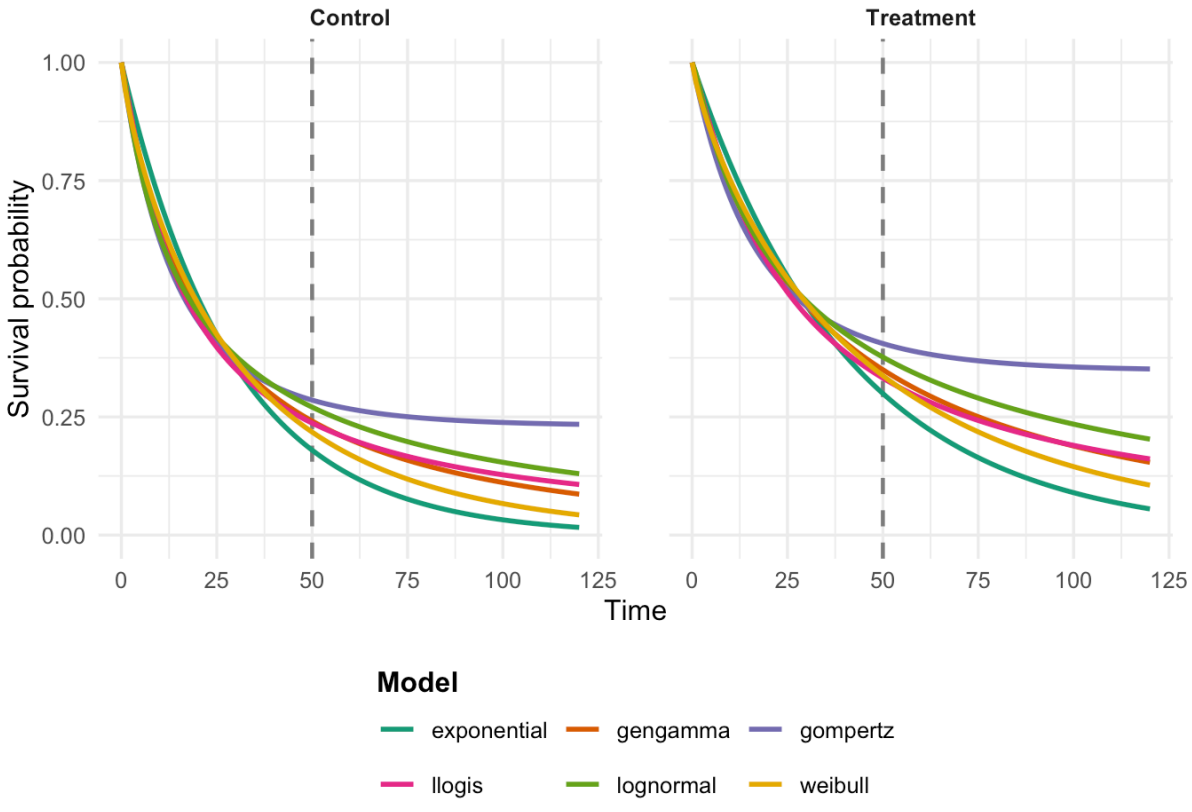
```
head(pr_long, 10)
```

```
## # A tibble: 10 × 6
##     time   est   lcl   ucl group   model
##    <int> <dbl> <dbl> <dbl> <chr>   <chr>
##  1     0 1     1     1     Control exponential
##  2     1 0.966 0.960 0.971 Control exponential
##  3     2 0.934 0.921 0.944 Control exponential
##  4     3 0.902 0.884 0.916 Control exponential
##  5     4 0.872 0.849 0.890 Control exponential
##  6     5 0.842 0.814 0.865 Control exponential
##  7     6 0.814 0.782 0.840 Control exponential
##  8     7 0.786 0.750 0.816 Control exponential
##  9     8 0.760 0.720 0.793 Control exponential
## 10     9 0.734 0.691 0.770 Control exponential
```

Hide

```
ggplot(pr_long, aes(x = time,
                    y = est,
                    color = model,
                    group = model)) +
  geom_vline(
    xintercept = 50,
    color = "grey50",
    linetype = "dashed",
    linewidth = 0.8
  ) +
  geom_line(linewidth = 1) +
  facet_wrap(
```

```
    ~ group,
    nrow = 1) +
  scale_color_brewer(
    palette = "Dark2",
    name = "Model"
  ) +
  scale_y_continuous(
    limits = c(0, 1),
    breaks = seq(0, 1, 0.25)
  ) +
  labs(
    x = "Time",
    y = "Survival probability"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    legend.position = "bottom",
    legend.title.position = "top",
    legend.title = element_text(face = "bold"),
    strip.text = element_text(face = "bold"),
    panel.spacing = grid::unit(1, "cm")
  ) +
  guides(
    color = guide_legend(
      nrow = 2,
      byrow = TRUE
    )
  )
```



Do not consider extrapolated curves validated solely because they fit the trial data. For each plausible model, report long-term survival and hazard predictions at decision-relevant time points, and compare them with relevant registries, longer-term studies, background mortality, biological expectations, and clinical-expert judgment. Discuss the comparability and limitations of each external source.

The base-case extrapolation assumes constant treatment effect throughout the period of interest, but this assumption should be critically evaluated in light of treatment-effect duration. Because post-trial relative effects are uncertain, evaluate at least three scenarios: treatment effect stops at the end of observed follow-up, wanes over a clinically justified period, or persists over the lifetime horizon. Inform the assumptions with observed hazards, mechanism of action, and clinical advice.

Step 5.Model selection & validation summary

Step 5.1 Model selection rationale

Model	Hazard / structural evidence	KM graphical fit	Q-Q diagnostic	Cox-Snell diagnostic	AIC; ΔAIC	External plausibility	Provisional conclusion
Exponential	Constant-hazard assumption not convincingly supported	Less favorable than Gompertz, log-normal, and log-logistic	Moderate deviation, particularly during mid-to-late follow-up	Less favorable than leading models; arm-specific departures observed	2215.113; 41.482	Not assessed	**Not preferred based on internal fit**
Weibull	Monotonic hazard structure potentially plausible	Less favorable than Gompertz, log-normal, and log-logistic	Moderate deviation, particularly during mid-to-late follow-up	Less favorable than leading models	2204.574; 30.943	Not assessed	**Structurally plausible, but with relatively weak internal-fit support**
Gompertz	Monotonic hazard structure potentially plausible	Best or among the best graphical fits	Not evaluated due to GofCens limitation	Better fit in the control arm and best fit in the treatment arm	**2173.631; 0.000**	Not assessed	**Provisional preferred model based on internal fit only**

Model	Hazard / structural evidence	KM graphical fit	Q-Q diagnostic	Cox-Snell diagnostic	AIC; ΔAIC	External plausibility	Provisional conclusion
Log-normal	Not assessed in the hazard-structure analysis	Among the better graphical fits	Moderate deviation, particularly during mid-to-late follow-up	Better fit in the control arm; treatment-arm fit inferior to Gompertz	2198.150; 24.519	Not assessed	**Potential alternative, but with weaker overall internal-fit support**
Log-logistic	Not assessed in the hazard-structure analysis	Among the better graphical fits	Moderate deviation, particularly during mid-to-late follow-up	Better fit in the control arm; treatment-arm fit inferior to Gompertz	**2179.306; 5.675**	Not assessed	**Principal alternative for structural sensitivity analysis, subject to long-term plausibility**
Generalized gamma	Not assessed in the hazard-structure analysis	Among the better graphical fits in later KM comparisons	Not evaluated	Not evaluated	2191.250; 17.619	Not assessed	**Evidence incomplete; further diagnostics required before assigning a scenario role**

Model selection should not be based on a single diagnostic. The provisional conclusions above synthesize structural assumptions, graphical fit, diagnostic plots, and information criteria. Long-term external plausibility was not assessed at this stage and should be considered before final model selection.

Step 5.2 Structural uncertainty and scenario analysis

Use the most clinically credible model as the base case, but retain other plausible distributions and modeling structures as scenarios. Scenarios should vary the survival distribution, PH versus arm-specific/time-varying effects, treatment-effect duration, and—where indicated—piecewise or flexible modeling. Report the effect on extrapolated mean survival and cost-effectiveness outcomes; parameter uncertainty within each scenario should be represented probabilistically.

Author biography

Xin (David) Zhao, PhD integrates epidemiology, evidence synthesis, quantitative modeling, and health economics to develop transparent, reproducible analytics that support healthcare and biotech decision-making.